How to run Spire

Primary entry point: `scripts/run_graph.py` (the LangGraph pipeline).

Commands below are marked:

- ◈ **ONE-TIME** — run once during setup
- ◈ **EVERY TIME** — run each session

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1. Setup

◈ ONE-TIME

```
python -m venv ~/venv/spire
source ~/venv/spire/bin/activate
pip install -r requirements.txt
pip install -e .
```

> `pip install -e .` is required — code lives under `src/spire/`, not the project root. Without it: `ModuleNotFoundError: No module named 'spire'`.

Credentials

◈ ONE-TIME

```
cp personas.example.yaml personas.yaml
```

```yaml
default_dev:
  fusion:
    base_url: "https://<your-pod>.oraclepdemos.com"
    username: "<fusion-user>"
    password: "<fusion-pass>"
    auth_type: "basic"
  oracle:
    dsn: "localhost:1521/FREEPDB1"
    username: "spire"
    password: "<db-pass>"
```

> ⚠ **`personas.yaml` is the single source for all pod and DB credentials. Update it whenever a password, pod URL or DB user changes** — a rotated Fusion password or a rebuilt DB container will otherwise fail at the first call with `ORA-01017` or a 401, with nothing else changed. It is gitignored, so nobody else's copy updates with yours.

Create the Oracle DB user:

ONE-TIME

```
docker exec -i oracle26ai_db sqlplus sys/Abcd1234@localhost:1521/FREEPDB1 as sysdba <<'EOF
CREATE USER spire IDENTIFIED BY "<db-pass>";
GRANT CONNECT, RESOURCE, CREATE VIEW, UNLIMITED TABLESPACE TO spire;
EXIT;
EOF
```

> Use `sys ... as sysdba`, not `pdbadmin` (`ORA-01031` on this image). `IDENTIFIED BY` must match `oracle.password` in `personas.yaml`.

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2. LLM keys

Provider via `SPIRE_LLM_PROVIDER` (default `gemini`), model via `SPIRE_LLM_MODEL`. Each run prints `[llm] provider=… model=…`.

Defaults live in `src/spire/config.py`:

```
GEMINI_MODEL = "gemini-2.5-flash"
ANTHROPIC_MODEL = "claude-sonnet-4-6"
OPENAI_MODEL = "gpt-5"
```

> `2.5-flash` not `2.5-pro`: a fresh `GEMINI_API_KEY` commonly has a **0 free-tier quota** for `2.5-pro` (`429 RESOURCE_EXHAUSTED, limit: 0`).

Gemini — public API key:

ONE-TIME per machine

```
export GEMINI_API_KEY="..."
```

or Vertex AI:

```
export GOOGLE_GENAI_USE_VERTEXAI=TRUE
export GOOGLE_CLOUD_PROJECT=<project-id>
export GOOGLE_CLOUD_LOCATION=us-central1
gcloud auth application-default login
gcloud auth application-default set-quota-project <project-id>
```

OpenAI

```
export OPENAI_API_KEY="sk-..."
export SPIRE_LLM_PROVIDER=openai
```

Anthropic

```
export ANTHROPIC_API_KEY="sk-ant-..."
export SPIRE_LLM_PROVIDER=anthropic
```

> To change a default permanently, edit the constant in `config.py` — one file, one place.

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3. Run

◈ **EVERY TIME** — environment. These must match `personas.yaml`; if you change credentials, change both:

```
export IKS_FUSION_BASE_URL="https://fa-esll-dev37-saasfademo1.ds-fa.oraclepdemos.com"
export IKS_FUSION_USERNAME="<fusion-user>"
export IKS_FUSION_PASSWORD="<fusion-pass>"
```

◈ **EVERY TIME** — Knowledge Graph. Always rebuilt from scratch, never incrementally:

```
python scripts/rebuild_kg.py
```

Optional checks:

```
python scripts/print_kg.py
python scripts/print_kg.py --counts-only
```

Two terminals:

```
# Terminal 1 — MCP server (keep running)
python -m spire.mcp_server.server

# Terminal 2 — a prompt
python scripts/run_graph.py --prompt "List all suppliers in India"
```

> **Restart the MCP server after ANY change under `src/spire/mcp_server/`.**

Override provider or model for one command:

```
SPIRE_LLM_PROVIDER=openai python scripts/run_graph.py --prompt "..."
SPIRE_LLM_MODEL=gemini-2.5-pro python scripts/run_graph.py --prompt "..."
```

Exit codes: `0` ok · `2` node error · `4` refusal (out of scope).

> Don't print a raw exit code next to a refusal in a loop — a screen of `exit=4` reads as a screen of crashes.

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4. Diagnostic flags

Default is quiet. `stdout` carries the answer; `stderr` carries everything else.

```
python scripts/run_graph.py --prompt "..." 2>/dev/null    # answer alone
```

Identical flags on `run_graph.py` and `run_prompt_set.py`.

Flag	Shows
`--show-parser`	parsed intent, KG menu size, refusal detail
`--show-planner`	selected plan, tool, arguments, filters
`--show-raw-output`	executor output per step
`--show-synthesizer`	response object prior to rendering
`--show-tokens`	per-call and end-of-run token counts
`--show-timings`	per-stage latency
`--show-all`	enable every flag above

Every flag except `--show-all` has a `--no-` twin:

```
python scripts/run_graph.py --prompt "..." --show-all --no-show-tokens
```

Progress lines and the summary footer are **never gated** — both on `stderr`, so `stdout` is unaffected.

The root logger is never touched (langchain, httpx, oracledb and fastmcp all log at DEBUG).

Precedence

Later beats earlier:

```
defaults  →  pyproject.toml  →  environment  →  command line
```

Committed default:

```toml
[tool.spire.diagnostics]
show_planner = true
```

Per-machine default:

```
export SPIRE_SHOW_PLANNER=1
```

Invalid flag names are reported, not silently ignored.

> `show_all = true` is a **no-op** in a file that already spells out every key. Set the individual keys, or use `--show-all` on the CLI.

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5. Prompt sets

```
python scripts/run_prompt_set.py --prompt-file scripts/prompts/00_dipstick.txt
```

Option	Effect
`--prompt-file FILE [FILE ...]`	one or more files, in the order given. Globs work: `scripts/prompts/0*.txt`
`--prompts "..." ["..."]`	inline prompts, overriding the set
`--all`	every legacy `prompt_set_*.txt` in the **scripts** dir — not `scripts/prompts/`. Mutually exclusive with `--prompt-file`.

The sets

Counts are runnable prompts (comments and blanks excluded), measured 2026-07-31.

File	n	Purpose	Baseline
`00_dipstick.txt`	6	depth-ordered triage, fast. **Run first** — the first failure tells you the depth at which the pipeline breaks	—
`01_supplier.txt`	40	supplier-rooted, incl. paraphrase variety and collision traps	—
`02_purchase_order.txt`	9	PO-rooted (P2, P3, P3b, P5, P6c)	—
`03_ar.txt`	16	accounts receivable	—
`04_columns.txt`	8	column selection — verification set since the feature shipped	—
`05_refusals.txt`	9	out-of-scope, consolidated. Fast, no pod call. **A refusal is a pass** — what matters is *how* it refuses	9/9
`06_slow.txt`	15	edge filters + aggregations. **Minutes each** — run overnight	—
`07_alacarte.txt`	12	column-feature regression (table, address flatten, graceful overrun)	12/12
`08_lifecycle.txt`	31	PO lifecycle regression. The five delivery prompts **must answer** — a refusal there is a failure	31/31
`10_refusal_examples.txt`	18	every example Spire promises in a refusal. **All must answer** — a refusal here is a bug in	18/18

File	n	Purpose	Baseline
		`capability.py`	
`qa_testcases.txt`	34	captured external QA prompts, verbatim, with per-prompt status labels. Not a regression set	—

Two fast sets worth running first: `00_dipstick` samples the whole pipeline by depth; `07_alacarte` drills the column-output feature.

> ⚠ **`scripts/prompts/scorer/parser_stress.txt` is a different format** (`EXPECT ||| prompt`) and needs the scorer. Feeding it to `run_prompt_set.py` sends `filter:status |||` to the parser as part of the prompt. It lives in a subfolder for that reason.

> `scripts/prompts/0*.txt` picks up `00` – `08` but **not** `10` or `qa_testcases` .

> `scripts/prompts/README.md` is stale — it predates `08` , `10` and `qa_testcases` , and five of its eight counts disagree with the files.

Reading the results

> **The tally is not a verdict.** `N/N clean` means nothing errored, not that the answers are right. `run_prompt_set.py` has reported "38/38 ran cleanly" while prompts were being wrongly refused, and `05_refusals` once reported 9/9 with three messages wrong.

Score each prompt into pass · fail · **wrong but did not error**. The third bucket is the one that matters. Check specifically for:

- an empty data array behind confident narrative prose
- a page count stated as a total ("25 suppliers" where 421 exist)
- claims about fields absent from the displayed projection
- arithmetic in the narration that is not in the rows
- customer-scoped results that never resolve an `AccountNumber`
- a generalisation true of the first few rows but not the set

Batch runs

Full run excluding the slow set:

```
for s in 00_dipstick 01_supplier 02_purchase_order 03_ar 04_columns 05_refusals 07_alacar
  echo "########## $s ##########"
  python scripts/run_prompt_set.py --prompt-file scripts/prompts/$s.txt
done 2>&1 | tee run_$(date +%Y%m%d).txt
```

Everything, overnight:

```
nohup bash -c '
for s in 00_dipstick 01_supplier 02_purchase_order 03_ar 04_columns 05_refusals 07_alacar
  echo "########## $s ##########"
  python scripts/run_prompt_set.py --prompt-file scripts/prompts/$s.txt
done' > overnight_$(date +%Y%m%d).txt 2>&1 &
```

The MCP server must stay up for the duration. If it is in a foreground terminal you plan to close, background it too.

Flag surface regression (only when the flag surface changes):

```
bash scripts/flag_matrix.sh      # ~17 min, 67 LLM calls
```

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6. Example prompts

Return data:

```
python scripts/run_graph.py --prompt "List all suppliers in India"
python scripts/run_graph.py --prompt "US suppliers with no invoices in the last 6 months"
python scripts/run_graph.py --prompt "Suppliers in the US with open purchase orders and n
```

Aggregation (slow — paginates the full range):

```
python scripts/run_graph.py --prompt "Top 10 suppliers by spend in 2022"
```

Refused:

```
python scripts/run_graph.py --prompt "Show me contract renewals due next quarter"
python scripts/run_graph.py --prompt "Top 10 suppliers by total spend"     # no date rang
```

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7. Tests

```
python -m pytest -q                  # 762 passed, 18 deselected, 1 xfailed (~2.5 min)
python -m pytest -q -m integration   # needs MCP server + KG + a provider key
```

> The 18 deselections are deliberate — `addopts = "-m 'not integration'"` in `pyproject.toml`. If that count changes unexpectedly, something in `pyproject.toml` was overwritten.

Troubleshooting

- **`ModuleNotFoundError: No module named 'spire'`** — `pip install -e .`
- **`ORA-01017`** — wrong DB password in `personas.yaml`, or the `spire` user doesn't exist (§1).
- **`ORA-01031`** creating the user — use `sys ... as sysdba`.
- **`401` from the pod** — Fusion password rotated; update `personas.yaml` and the `IKS_FUSION_*` exports.
- **`429 RESOURCE_EXHAUSTED` / `limit: 0`** — zero free-tier quota for that model. Use `gemini-2.5-flash` or switch provider.
- **`429 / credits depleted`** — provider out of credit.
- **`404 not_found: model ...`** — bad `SPIRE_LLM_MODEL`.
- **`DefaultCredentialsError`** — `gcloud auth application-default login`.
- **`No Oracle DB credentials`** — `personas.yaml` oracle block missing

and no `SPIRE_DB_*` env vars.

- **MCP connection errors** — server not running, or not restarted after an `mcp_server/` change.
- **Diagnostics print above the answer when piped** — fixed; `diagnostics.answer()` flushes.
- **A whole set refuses with a short entity list** — the KG is stale relative to the code. Run `scripts/rebuild_kg.py` before reading anything into the results.

Legacy

```
# ask.py — Track A LangGraph agent (1/7 baseline; cleared for removal)
python scripts/ask.py "Show invoices for Lee Supplies after January 2024"
```

> `run_plan.py` was deprecated and deleted. Use `run_graph.py`.